EEG-to-Image Cross-Subject Decoding: A Decision-Path Research Journal

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Nature of this document: Not a history sheet. This traces *what we found, how each finding changed the next decision, and where those forks led*, so the reasoning path is legible, not just the outcomes. Where a decision was a genuine fork (advisor call, data reality, a number that came back wrong), it is marked as such.

The spine of the whole project

One question runs through everything:

> **When an EEG-to-image decoder learns, does it learn the *subject* or the *task*?**

Every fork in this journal is a consequence of chasing that question: first proving the answer for retrieval (Paper 1), then discovering that *where* you can even ask it is sharply constrained by data (the second-dataset saga), then asking whether the answer is specific to retrieval or general across tasks (Paper 2). A second, quieter thesis emerged along the way and deserves to be stated up front because the data keeps arguing for it:

> **Retention, the fraction of in-domain performance that survives a subject change, is the metric this whole subfield should be reporting, and the classification probe is where that argument gets its cleanest test.**

Phase 0 — Getting a trustworthy baseline (June 29 – July 1)

Where we started. A working THINGS-EEG2 pipeline (the dongyangli-del / ATM repo) existed but had never had its encoder weights adjusted properly, so it generated a hallucinated image. The first instinct was to retrain from scratch. The first *decision* reversed that: inspecting the artifacts showed the baseline was already producing valid $[200, 1024]$ features, so **retraining the encoder was unnecessary**; the hallucination was downstream (guidance/normalization), not the encoder. This set the tone for everything after: **inspect before you rebuild.**

The environment reality. Plans shifted from a multi-GPU lab machine and a shared cluster to the actual setup: a Windows desktop with an RTX 4080, Git Bash, conda env `BCI`. ARCC (University of Wyoming, H100s) was raised early as the eventual scale-out but was repeatedly not ready, so the discipline became: **verify and gate locally on the 4080, ship verified units to ARCC later.**

Why this mattered later. The inspect-first reflex is the same move that, weeks later, caught a commented-out encoder load, a microvolt scaling bug, and a single-trial dataset, each of which would have produced confident, wrong results.

Phase 1 — The finding takes shape (July 1)

The pivotal run. LaBraM (foundation encoder) was integrated as a swappable encoder and run intra-subject against ATMS (specialist). The specialist won intra-subject, and the reaction captured the whole insight before any statistics existed: a specialized encoder built from the ground up can be more efficient than one that knows everything and has to dissect it all.

The fork that made it a paper. The instinct could have stopped at "ATMS wins intra." Instead the research question was explicitly redirected to **cross-subject variability**: the real interest was not reconstruction quality but *the degradation between subjects.* That reframing is the entire reason there is a paper: the story is not who wins in-domain, it is *who collapses when the subject changes.*

A dead end that clarified the direction: LoRA. Considerable effort went into LoRA-fine-tuning LaBraM. It surfaced a genuine silent failure (`LoRA A grad is None`, grad norm 0.0: the adapters were not training while the projection head was). Decision after debugging: **abandon LoRA, use full fine-tuning.** A documented negative that kept the comparison clean (both encoders fully trained).

The reconstruction sub-thread. The pipeline was validated with zero-shot reconstruction (sub-08, CLIP ∼0.80), but the reconstructions never matched the ATM paper's fidelity. The important realization: the earliest bad images came from an *untrained* encoder (the commented-out load), and reconstruction was never the point; it validates that the embeddings capture real visual content. Decision: **reconstruction stays as validation only; the paper's claim lives in retrieval retention.**

Phase 2 — Making it airtight (July 1 – July 9)

The 10-fold sweeps. Both encoders were run across all ten subjects, intra and LOSO. This is where retention first became the load-bearing metric: ATMS intra ∼0.32, LOSO ∼0.11 (∼33% retention); LaBraM intra ∼0.12, LOSO ∼0.11 (∼92% retention).

The subtlety that had to be got right. On *raw* LOSO accuracy the two encoders are a **statistical tie** (ATMS 0.116 vs LaBraM 0.105; paired $t = 1.09$, $p = 0.30$). It would have been tempting, and wrong, to claim "foundation decodes better cross-subject." So the claim was pinned to the **degradation asymmetry**, not a raw ranking. This is the single most important interpretive decision in the paper; getting it wrong would have made the work refutable in one line.

Replication as deliberate hardening. The asymmetry was replicated across three seeds:

Seed	ATMS retention	LaBraM retention	Paired t	Subjects ATMS worse
42	36%	89%	11.0	10/10
43	33%	92%	13.67	10/10
44	31%	94%	9.00	10/10
3-seed mean	**33.3% ± 2.4**	**91.5% ± 2.7**	9.0–13.7	30/30

Wilcoxon $W = 0$ throughout. Running seeds 43 and 44 (cheap, ∼overnight each) converted an observation into a result no reviewer can wave off as a lucky split.

The citation-verification episode. While fact-checking the bibliography, several 2025–2026 papers were initially treated as *possibly fabricated* because they were unrecognizable. Web search confirmed all of them real; they simply postdate the training cutoff. The lesson, applied ever after: **not recognizing a citation is not evidence it is fake; for recent work, search, don't doubt.** This also sharpened the novelty claim: prior work already fine-tunes LaBraM for EEG-to-image, so the contribution narrowed, honestly, to *isolating the specialized-vs-foundation degradation asymmetry with paired statistics, pipeline held fixed.*

Advisor fork #1 (methodology). The advisor initially disliked the v2–v200 retrieval methodology and wanted generated-image comparisons. The line was held correctly: v-way retrieval *is* the field-standard measure, and generating images would look worse precisely when the result is actually good. Decision: **keep retrieval; clarify the methodology rather than switch to a weaker measure.**

Phase 3 — The second-dataset saga: where the question can even be asked

Each dataset failed differently and each failure changed the next decision. The advisor **mandated a second dataset** (July 9), which forced the exploration that ultimately produced a finding of its own.

Fork: which second dataset? Two *kinds* of replication were distinguished: new-images/new-rig/new-concepts (EEG-ImageNet), a strong generalization test that changes everything at once; versus new-rig/same-stimulus (THINGS-EEG1), a cleaner but more modest replication. **Advisor fork #2:** she chose EEG-ImageNet, the harder path.

EEG-ImageNet — the empirical negative that reframed the project. The integration was a gauntlet of silent-failure bugs: a per-synset sourcing insight collapsed a feared 1.3 TB download to ∼7.4 GB; ∼496 of 4000 images were missing (an earlier ImageNet-21k snapshot, confirmed by the dataset author), handled by keeping 70 categories (3191 images); a decisive **microvolt voltage-scale bug** (values ∼1e-6 silently killing learning, fixed by per-channel z-scoring); and channel handling hard-verified against LaBraM's `standard_1020` vocabulary. The result, across four configurations (ATMS intra, LaBraM intra with train acc →0.96, LaBraM LOSO with ∼15× the data, and an image-level diagnostic), was **chance-level held-out retrieval every time.** The encoders learned the training set and did not generalize. This reframed the whole arc: the problem was EEG-ImageNet's weak per-trial signal, not the pipeline, which is exactly what made THINGS-EEG2 special.

MSS — the structural negative, ruled out before compute. Reading the preprocessing script (no training at all) showed the low-speed paradigm is **single-trial with category-only**

labels (20 classes): image-level retrieval is architecturally impossible, there is no per-image target. **Decision:** document MSS as a structural negative. The strongest kind of negative, determined from the data's design, costing hours instead of a weekend.

Alljoined — the empirical negative, pushed to certainty. Alljoined (8 subjects, 64ch BioSemi, 960 COCO images, ∼7.9 reps/image) was built completely from scratch: Parquet loader, per-channel z-score, FFT-resample 334→250, CLIP ViT-H-14 features (960/960, 0 missing), montage validated name-by-name. Across four conditions (including 8-subject pooling to ∼8× the data), **held-out retrieval stayed pinned to its chance floor while train accuracy climbed to 0.97.** Because ATMS ignores channel names and *also* hit chance, the negative cannot be a montage artifact: it is a property of Alljoined's per-trial signal.

THINGS-EEG1 and THINGS-MEG — ruled out before wasting effort. THINGS-EEG1 is **single-presentation** (each image seen once), corrected by web search *before any download*, the same structural wall as MSS. THINGS-MEG genuinely supports retrieval but has **only 4 subjects** (a hard ceiling) and is cross-modality (271 MEG channels); parked as optional corroboration.

The finding hiding in the negatives.

Dataset	Repeats?	Image labels?	Verdict / failure mode
THINGS-EEG2	Yes (4–80×)	Yes	**Works** (primary)
THINGS-MEG	Yes (12×)	Yes	Works; 4 subjects, cross-modality
THINGS-EEG1	No (single)	Yes	Chance (structural): single-presentation
EEG-ImageNet	Partial	Yes	Chance (empirical): weak per-trial signal
MSS	No (single)	No	Impossible: category-only labels
Alljoined1	Yes (∼7.9×)	Yes	Chance (empirical): weak even pooled

Image-level EEG-to-image retrieval is currently concept-space-bound to the THINGS family. That is a real contribution: it tells the next researcher exactly which datasets can carry this task and why the others cannot.

Phase 4 — Locking Paper 1 (July 13)

Advisor fork #3. The advisor judged the paper nearly done and asked to **finish the Discussion and Conclusion and lock it as Paper 1**, with the model expansion becoming a second paper, and to mention as much as possible in this first paper for the field's benefit. So the EEG-ImageNet negative stays in, framed honestly as the concept-space-bound signal. A factual error (an RSVP mis-attribution) was caught at lock-time by web search. **Paper 1 status:** locked, IEEE conference format, `root.tex`.

Phase 5 — Paper 2, and where retention becomes the argument (July 14)

The idea. Is the asymmetry *task-general*, does it show up in classification, not just retrieval? This matters twice: it would promote the finding from a property of retrieval to a property of the representation; and **classification is supported by the non-THINGS datasets that cannot do retrieval**, so a classification version could finally test concept-generalization on genuinely different content.

The probe. Rather than retrain, the probe reuses the 10 intra-trained THINGS checkpoints: for each subject's frozen intra encoder, extract embeddings on all subjects, fit a linear probe, measure within- vs cross-subject concept accuracy. A design bug was caught in the smoke test: the THINGS *test* set has one image per concept, so it cannot support classification; switched to the training set (10 images/concept), capped at 100 concepts.

Committing the metrics before the run. The first (2-subject) signal was weak and confounded (the specialist starts higher within-subject, distorting a naive retention ratio). The response was to commit, *before the powered run*, to three fairer metrics and report all three regardless of outcome (triangulation, not p-hacking):

1. **Chance-corrected retention** = (cross − chance)/(within − chance).
2. **Normalized drop** = (within − cross)/(within − chance).
3. **Rank-based transfer** = correlation of per-concept accuracy within vs cross (does the *structure* transfer, independent of magnitude).

Phase 6 — The probe settles, the panel widens, the code ships (July 15–21)

The transfer probe completed, and it was reported whichever way it fell. The full ten-subject, three-metric run resolved the task-generality question, and the answer is a nuanced one rather than a clean "yes":

Encoder	Chance-corr. retention R	Normalized drop D	Rank transfer ρ
ATMS (specialized)	0.154	0.846	0.083
LaBraM (foundation)	**0.234**	**0.766**	0.084
Paired t (n=10)	3.15	3.15	0.06
p (two-sided)	**0.012**	**0.012**	0.95
LaBraM better in	9/10	9/10	5/10

The retention asymmetry *is* significant (p = 0.012, nine of ten subjects), but far weaker than retrieval's (an ∼8-point chance-corrected gap versus ∼60 in retrieval), and **rank transfer is null** (p = 0.95): neither encoder preserves the *structure* of which concepts survive. The honest reading is that subject-invariance and task-strength are **separable** properties, "strong in retrieval, weaker in classification," not a clean task-general claim. One point sharpened the retention argument: on raw within-subject accuracy the specialist scores *higher* (0.116 vs 0.067, p = 0.0001), and only chance-corrected retention reveals the foundation encoder as the one that transfers. As in Paper 1's raw-LOSO tie, raw accuracy points the wrong way and retention points the right way, now demonstrated on a second task.

The second foundation model, integrated and verified. To turn the two-model observation into a two-cluster *principle*, CBraMod (6.97M) was added beside LaBraM (6.45M). The decisive discipline was refusing to trust it blind: CBraMod was taken from its *official* repository (braindecode ships an architecture-only version whose weights would have been silently random), and the pretrained weights were verified genuinely loaded, 100% key match, 208/209 parameters differing from a random reinitialization. This closed the single largest silent-failure risk a foundation model carries. Integration also surfaced and fixed three real bugs: a hard import that took down the entire encoder registry, a `models/` namespace collision between CBraMod's package and the project's, and a missing `logit_scale/loss_func` contract that the retrieval engine requires. The two foundation encoders are deliberately matched in size, so foundation status is not confounded with parameter count; the specialist panel (EEGNet 1.19M, EEGConformer 0.64M, ShallowFBCSPNet 0.89M, ATMS 3.20M) spans a range that overlaps them, so a clean cluster split, if it emerges, cannot be attributed to capacity alone.

The epoch-budget finding (a canary that paid for itself). The panel was nearly shipped at a fixed 40-epoch budget, the budget that converged ATMS and LaBraM. A single-encoder canary run caught that this **severely undertrained EEGConformer**: v200 top-1 was 0.010 at 40 epochs versus 0.075 at 200 (early stopping first fired at epoch 85; true convergence ~196). Shipping at 40 would have reported undertraining as "specialist collapse" and invalidated the panel. The protocol was changed to a **300-epoch cap with early stopping**, so every encoder trains to its own convergence (ATMS ~40, LaBraM ~22, EEGConformer ~200) with no hidden per-encoder tuning. Verification of the locked Paper 1 runs confirmed both its encoders had genuinely converged inside 40 epochs, so Paper 1 is unaffected. A related dissociation emerged from the loss curves: LaBraM has a *larger* train/val gap than ATMS yet retains far more across subjects, so within-subject memorization and cross-subject overfitting are separable, which falsified a tidy mechanistic hypothesis and is more interesting than the story it replaced.

The LOSO invocation bug. The sweep's leave-one-subject-out folds crashed with an empty training set. The cause was passing a single subject to `--mode loso` (training on that subject minus itself). The fix was not new code: `train_unified.py` already had a `--test_subjects` argument that isolates one fold correctly (train on the pool minus the held-out subject). An invented patch was discarded in favor of the machinery that already existed, verified on a real fold (132,320 trials from the correct nine subjects).

Repository organization and release. The working tree had accumulated ~36 stray files (backups, logs, dead-end patches from the concluded EEG-ImageNet/Alljoined negatives). A conservative declutter script archived them into `_archive/` with a protected-file guard, and the verification gates stayed green afterward. The code was then pushed to GitHub, cleanly: an upload guard (a `.gitignore` verified in practice) **caught a 297 GB nested data repository, with its own 149 GB of Git-LFS objects, before it could be committed**, and the final push was 430 KB of code and documents with no data or weights. A README and a reproduction guide were written to match the repository's real layout.

Compute. Local development, the intra-subject sweeps, and the transfer probe all ran on the RTX 4080. The leave-one-subject-out folds are the parallel-execution case: EEGConformer LOSO alone is on the order of 130 GPU-hours run serially, which a Slurm array on ARCC (or NCAR) reduces to about the wall time of a single fold. This is now a concrete, quantified reason for the cluster rather than a vague preference, and the decision between ARCC (H100s) and NCAR (A100s) remains with the advisor.

Phase 7 — The cluster, and the panel becomes a three-seed result (July 22–26)

Why we moved. The panel's leave-one-subject-out folds each train on nine subjects, so they cost roughly nine times an intra-subject run per epoch. Run serially on the one RTX 4080, the panel's LOSO sweeps across three seeds would have taken weeks. The decision, deferred at the end of Phase 6, was made: migrate to the University of Wyoming ARCC MedicineBow cluster (H100 and

A30 partitions, Slurm array scheduling). The move turns a multi-week serial grind into roughly the wall time of the slowest fold by fanning folds out across nodes.

The staging was a saga, and every step of it was a silent failure caught by a gate. Nothing about moving the pipeline to a new machine was trusted; each assumption was checked, and the checks kept finding things that would have run to plausible-looking numbers on broken foundations. In rough order: three source directories (the model package, the LaBraM scaffold, and the LaBraM repository itself) were being excluded from version control by `.gitignore` rules written to keep weights out, so they never reached the cluster; the LaBraM repository had been committed as an empty submodule pointer rather than its files, and had to be cloned directly on the cluster where a checksum confirmed the pretrained weights were byte-identical to the reference; the braindecode wheel that installed was an incomplete 0.8 upload missing an entire subpackage, distinct from the 0.8.1 the code needs; two pip packages were simply absent; the repository and data paths inherited the development machine's folder names and pointed at nothing; and, most insidiously, `conda activate` silently no-opped in batch context and left the job running on the base interpreter with no PyTorch at all, which was fixed by sourcing the conda profile explicitly and adding a hard guard that exits the job if the wrong Python or a missing Torch is detected. Finally an out-of-memory kill during nine-subject data assembly (the held-out pool is concatenated into one large tensor, briefly doubling memory) was resolved not by guessing but by computing the true peak from the tensor shapes and setting the allocation with headroom. Six independent verification gates were built and had to go green before a single real training job was allowed to run. Every one of these would have produced either a wall of failed array tasks or, worse, a foundation model quietly training on random weights, the exact failure the whole discipline exists to prevent.

The panel ran, and it replicated. With the gates green, the full panel trained across three seeds (42, 43, 44), each run stamped with its git commit, resolved arguments, library versions, and seed so that cross-seed and cross-partition comparisons are auditable rather than trusted. The result held across all three seeds without a single reversal, and three findings crystallized. *First, the size confound closed:* the three specialists span a five-fold range in parameters (EEGConformer 0.64M, EEGNet 1.19M, ATMS 3.20M) and all retain between roughly 24% and 40%, so capacity does not predict retention within the class. EEGNet is the sharpest case, smaller than either foundation model yet collapsing to ∼37%. *Second, and most interesting, the foundation class is not uniform:* CBraMod retains ∼54% and LaBraM ∼92%, a within-class spread of ∼39 points that exceeds the ∼21-point gap between the worst foundation encoder and the best specialist. The clean slogan "foundation encoders hold" turned out to be too strong; the honest and more useful claim is that pretraining reliably buys *some* subject-invariance whose magnitude varies by model and pretraining corpus. *Third*, CBraMod, the encoder carrying that non-uniformity claim, was confirmed at all three seeds (61.1/53.9/47.8%), so its intermediate position is real and not a single-draw artifact. A fourth specialist, ShallowFBCSPNet, was run and then excluded on principle: its within-subject retrieval sat at 0.018, near the 0.005 chance floor, so any retention ratio built on it would be noise over noise. Saying so plainly, and not running its LOSO folds, was the correct call.

Phase 8 — Two tasks, one mechanism (July 26–27)

The dissociation. The representation-transfer probe from Phase 6 was extended from two encoders to the full panel, freezing each intra-trained encoder and training a linear classifier on its features to measure how much *classification* skill survives across subjects. The panel run reproduced the published ATMS value exactly (chance-corrected retention 0.154), which validated the extended pipeline before any new row was interpreted, and then it delivered a genuine dissociation. Ranked by classification retention: LaBraM 0.234, EEGConformer 0.163, EEGNet 0.155, ATMS 0.154, and CBraMod *last* at 0.114, below every specialist. So in retrieval both foundation encoders beat every specialist, but in classification only LaBraM does. CBraMod's strong retrieval retention is therefore **readout-specific** rather than a subject-invariant representation, while LaBraM's invariance is **task-general**. The methodological lesson compounded: retrieval retention alone does not certify that an encoder has learned invariant features, so retention must be reported across tasks. And a second inversion echoed Paper 1's, LaBraM has the *lowest* raw within-subject classification accuracy (0.067) yet the highest chance-corrected retention, so raw accuracy pointed the wrong way for the third time across two tasks.

The mechanism. The panel establishes *that* the two foundation models differ; the natural question is *why* LaBraM's fine-tuning converges to strong invariance while CBraMod's does not. Two hypotheses make opposite predictions: either CBraMod's pretrained features are invariant and fine-tuning destroys them (freezing the backbone should then rescue it) or CBraMod's pretrained features are simply less invariant (freezing should not help). Implementing the test required a real code change, because the `freeze_backbone` argument in the LaBraM wrapper turned out to be dead code, accepted by the constructor and never used, a discovery that was itself a small vindication of reading the code rather than trusting the flag. A command-line switch was threaded through to both foundation wrappers, freezing only the backbone and leaving the projection head trainable so the frozen *features* are still mapped into CLIP space, and forty folds were run at seed 42.

The answer was unambiguous, and it is deliberately *not* a retention percentage. Freezing the backbone destroys the retrieval task for both foundation models: LaBraM's frozen within-subject retrieval collapses to 0.0085, about 1.7× the chance floor, and CBraMod's to 0.0655 against its full-fine-tune ∼0.13, with near-flat loss curves (training loss moved from 4.854 to 4.845 over nineteen epochs, validation loss essentially not at all). The retention ratios one could compute from those numbers are chance-over-chance and were set aside; the finding is the flat curve itself. Because freezing did *not* rescue CBraMod toward LaBraM, the "fine-tuning destroys good features" hypothesis is ruled out. The subject-invariance that matters is a product of fine-tuning, not a property of the frozen pretrained representation, and the two models differ in *where fine-tuning converges* from their respective starting points. The claim was scoped honestly to linear read-out: a frozen 200-dimensional representation into a linear CLIP projection is a hard test, so the precise statement is that frozen features do not *linearly* support the task.

Where it ends. With the mechanism in hand and the panel replicated, the decision was made to stop running experiments. The temptation, named plainly in the log, was to keep launching sweeps (a sixth encoder, a second dataset), and the discipline was to recognize that as journal-extension work rather than the REU and to turn to writing what is solid. The one remaining computation at the close was a single slow leave-one-subject-out fold finishing EEGConformer's third seed; every finding was already established without it.

Phase 9 — Retention without shared structure, and knowing when to stop (July 27–28)

The last stretch began as a rescue and turned into both a third dissociation and a lesson in scope. With cluster access closing, the seed-42 intra checkpoints for the panel encoders had to come off the machine before they were lost; a filtered copy, staged to a private repository and pulled back to the desktop, secured them, and the public test data was re-downloadable regardless. The checkpoints mattered because they enabled the question that had been waiting since the frozen-backbone result: if high retention does not require the frozen features, and does not guarantee task-generality, does it at least require that subjects share a representation? We measured it directly with centered kernel alignment.

Getting the measurement honest took more care than getting a number. Raw CKA is biased upward at two hundred concepts and the bias depends on dimension, so it went to a debiased estimator verified to read zero for independent inputs. Encoders differ in whether they normalize their output, so every representation was pushed into the same retrieval geometry, the exact tensor the retention metric already scores. The extraction point was checked against the retrieval engine line by line so that what CKA saw was what retention saw. Only then did the result mean anything, and it was the sharpest dissociation yet: the highest-retention encoder, LaBraM, had near-zero cross-subject similarity, two orders of magnitude below the others, while its representations remained individually rich. Retrieval retention does not require a shared cross-subject code; each subject's decoder recovers the target from an idiosyncratic geometry of its own.

The phase also held the clearest single instance of the discipline the whole project runs on. An early reading called LaBraM "rank-collapsed" on the strength of a two-subject spot check; the ten-subject mean showed its effective rank was mid-pack, and the collapse interpretation was wrong. Computing across the full panel rather than a convenient sample corrected it, and the corrected finding was stronger than the mistaken one. The same discipline excluded ATMS from the analysis: its checkpoints predated configuration-stamping and could not be provenance-verified, so they stayed out, even though including them would have completed the panel.

From the three dissociations came a small method. If retention and representational alignment are separable, an encoder should be characterized on both axes rather than ranked on one, and placing the four encoders on that plane produced a counterintuitive picture: the specialists, whose from-scratch codes are most similar across subjects, transfer worst, while LaBraM, least similar of all, transfers best. A single retention number ranks LaBraM first and hides the mechanism; only the two-axis view makes it legible.

The phase ended not with another experiment but with a scoping decision, which was the right kind of ending. A full journal had been the working ambition, and the honest read on the evidence was that the material did not support it: one dataset by the corpus boundary, a plane resting on four encoders, a seed-robustness test blocked because the foundation encoder's other-seed intra checkpoints were never retained, validation of the method requiring training runs of days. None of these were failures of effort; they were the shape of what the data could bear. The mature move was to let the two-encoder result stand as the conference paper it already is, carry the panel and the dissociations and the two-axis method into a workshop paper scoped as a diagnostic framework and a proof case, and leave the validation documented as the next paper rather than forced into this one. Knowing that a whole journal was out of scope, and stopping there deliberately, was itself a result: an accurate read on the size of the contribution, which is the thing a research project most easily gets wrong and this one, in the end, got right.

A coda, and the last instance of the spine. Closing out the paper turned up one more silent error, and it was in the headline numbers. Tracing every published figure back to the run directory that produced it — something never done before, because the fold data had always been assumed rather than located — showed first that a search which had concluded the specialist's per-subject folds were lost had in fact searched one of two results trees and truncated its own output; the folds were intact. With all sixty of them found, each seed's ten-fold means were recomputed and matched the published table exactly, which confirmed the fold selection. But the retention column did not survive the same test: seed 42 had been computed as a mean of per-subject ratios while seeds 43 and 44 were ratios of means, so a single table mixed two definitions and the three-seed mean averaged across the change. Per-subject ratios turned out to be the wrong choice in any case — near the chance floor individual foundation folds exceed 100% retention, and seed 43's per-subject mean is 100.3%. Retention was unified to the ratio of means throughout, which moved the foundation encoder's headline from 93.3% to 91.5% and cost nothing in the finding, and the per-subject spread was handed to a statistic built for it: the normalized drop D, the fraction of *above-chance* performance lost, which answers the baseline-dependence objection with a paired test rather than an argument (67.1% $\pm$ 2.4 vs 2.8% $\pm$ 3.6, significant in all three seeds). The lesson is the one the whole project runs on, arriving at the very end and aimed at its own conclusions: a number that reproduces is not the same as a number you can trace, and only the tracing tells you which formula produced it.

The methodological spine (the reason any of it is trustworthy)

Across every phase, the same discipline caught the same class of error: **"no error" is not "correct."** A partial catalog, all real:

- Commented-out encoder load (generation ran on random weights, no error).
- Microvolt voltage scale (right shape, wrong values, no learning), fixed by z-scoring on both EEG-ImageNet and Alljoined.
- LoRA adapters not training (`grad None`, norm 0.0) while the head trained.
- Channel-name aliasing to the CLS token (every montage validated against `standard_1020`).
- Config reaching the data but not every model layer (ATMS's hardcoded 250 / 1440).
- `val-loss-0.0000` early-stopping truncating runs and loading the wrong checkpoint.
- A foundation model that would have run on silently-random weights (caught by an explicit weight-realness check).
- An epoch budget that undertrained one encoder and would have misreported it as collapse (caught by a canary run).
- A 297 GB data repository about to be committed to git (caught by the upload guard).
- An entire pipeline's worth of cluster-staging failures, source directories excluded from git, a foundation repo committed as an empty pointer, an incomplete library wheel, and a conda activation silently falling back to a Torch-less interpreter (each caught by a staging gate before real GPU time).
- A `freeze_backbone` flag that was accepted and never used, so the frozen experiment required real code, not a switch (caught by reading the code, not trusting the argument).
- A comparability gate that refused to average a frozen run against a full-fine-tune run of the same encoder and seed, which would otherwise have produced a plausible but meaningless mixed table.
- Factual errors caught by web search before reaching the paper (RSVP attribution; THINGS-EEG1 single-trial; MEG's 4-subject ceiling; the "fabricated citation" scare).

Gate-first testing, proving a result at small scale before committing compute, repeatedly converted potential weekend-long mistakes into hour-long determinations, most starkly MSS (structural, no compute), THINGS-EEG1 (ruled out pre-download), and the EEGConformer epoch budget (one

GPU-hour to prevent a poisoned panel).

Where the paths stand now

- **Paper 1:** locked. The degradation asymmetry, three-seed, retention-framed, with the raw-LOSO tie stated honestly and the EEG-ImageNet negative kept in. Numbers harmonized to the three-seed means; citations verified; the cross-dataset section rewritten as a data-availability boundary. Pending only a compile with the real IEEE class before submission.
- **Dataset landscape:** fully mapped and itself a contribution; retrieval is concept-space-bound to THINGS, and the negatives are documented with distinct failure modes.
- **The panel:** complete and three-seed. Retention tracks class, not size (three specialists across a five-fold parameter range all retain 24–40%), and the foundation class is itself non-uniform (CBraMod ~54%, LaBraM ~93%), a more careful and more useful claim than a clean two-cluster split.
- **Cross-task dissociation:** the panel-wide classification probe shows CBraMod last despite mid-pack retrieval retention, so its advantage is readout-specific while LaBraM's is task-general. Retention must be reported across tasks.
- **Mechanism:** a frozen-backbone control establishes that the foundation subject-invariance is a fine-tuning phenomenon, not a property of the frozen pretrained features, and that freezing does not rescue CBraMod toward LaBraM.
- **Code:** decluttered, documented, and released to GitHub with data and weights safely excluded.
- **Compute:** migrated to ARCC MedicineBow (H100/A30), config-stamped and gated; the panel and both control experiments ran there across three seeds. The decision now is to stop running and write, treating a second dataset or a representational (CKA) analysis as journal-extension work rather than the REU.

The arc in one paragraph

We set out to ask whether an EEG-to-image decoder learns the subject or the task, and answered it cleanly for retrieval: specialists overfit and collapse cross-subject (~33% retention) while foundation encoders hold (~92%), a *degradation asymmetry* that raw accuracy hides because the two are tied on raw LOSO, which is precisely why retention is the metric that should be standard. Mandated to replicate on a second dataset, we discovered instead that the question can barely be asked anywhere else: of five datasets, only the THINGS family supports image-level retrieval, and each of the others fails for a distinct, now-documented reason. That mapping became a contribution of its own. Pressed on whether the asymmetry is a property of the task or of the representation, the classification probe returned a subtler answer than hoped, significant on retention but null on concept structure, establishing that subject-invariance and task-strength are separable. In parallel the comparison was widened to a full five-encoder panel, replicated across three seeds on a cluster whose every staging step turned out to hide a silent failure, and the panel returned a sharper thesis than the clean slogan it began with: retention tracks encoder class rather than size, but the foundation class is not uniform, so pretraining buys some subject-invariance whose amount depends on the model. A cross-task probe then showed that even strong retrieval retention need not mean an invariant representation, and a frozen-backbone control located the invariance in fine-tuning rather than in the pretrained features themselves. Throughout, the thing that made the results trustworthy was refusing to treat "no error" as "correct": every value verified against the data, every claim against a source, every scale-up gated on a small proof first, and, at the end, the discipline to stop running experiments and write down what was solid.